INCH-POUND

MIL-S-13949/23B
25 January 1993
SUPERSEDING
MIL-S-13949/23A
20 September 1988

MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, PREPREG, BASE MATERIAL BF
(NONWOVEN ARAMID REINFORCEMENT, EPOXY RESIN, FLAME RESISTANT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-13949.

REQUIREMENTS:

Reinforcement material: B for nonwoven aramid. All parameters shall be as detailed in table I.

Resin system: F in accordance with MIL-S-13949.

Special consideration (third letter designator): G or T in accordance with MIL-S-13949.

Woven reinforcement thread count: Not applicable.

Qualification inspection.

Inspection routine: The inspection routine shall include the permittivity and loss tangent tests in accordance with MIL-S-13949.

Extent of qualification: Qualification to third letter designator G or T does not extend to each other. Qualification of any reinforcement material style in table I extends to all other reinforcement styles in table I.

Conditioning of specimens prior to test: Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method.

Property values, all reinforcement styles.

Flammability, duration of burning time, each flame application: 15 seconds maximum.

Volatile content (conditioning of specimen prior to test, E-0.25/163): 1.5 percent maximum.

Electric strength, average: 1500 volts/mil minimum.

Loss tangent (at 1 MHz): .035 maximum.

Permittivity (at 1 MHz): 4.5 maximum.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

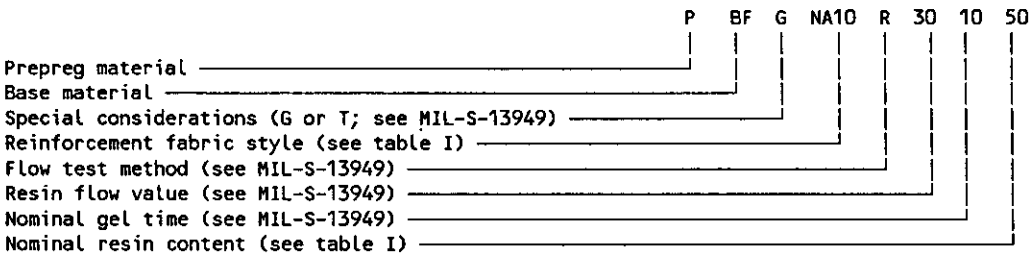
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TABLE I. Nonwoven aramid reinforcement construction and prepreg characteristics.

Reinforcement material		Prepreg selection data	
Style	Weight, ±15 percent (ounces per square yard)	Resin content range (percent)	Nominal thickness range, cured (inches/mm)
NA10	1.00	40 - 70	.0020 - .0030/(0.05 - 0.08)
NA16	1.60	40 - 70	.0025 - .0035/(0.06 - 0.09)
NA20	2.00	40 - 70	.0035 - .0050/(0.09 - 0.13)
NA25	2.50	40 - 70	.0040 - .0055/(0.10 - 0.14)

Supersession information: Not applicable.

Type designation: Consists of "PBF" prefix followed by coded alphas and numeric characters as shown in the following example:



Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
NSA - S5
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5998-0027-11)

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